

CSC215 Lab1 – Linux Warmup Exercise

Total: 100pts (see Blackboard DueDate)

What to turn in: zip and upload all your related files to **Assignment 1** in Blackboard. Your zip should include the following files:

- (1) CSC215Lab1.docx (with all questions completed)
- (2) vim.txt
- (3) req.txt
- (4) savepasswd.py

1. [VIM Practice. 10pt] Insert 100 lines of “assignment 1” into a file named “vim.txt”. Then jump to line 50, and change the “1” to 2. Then replace every “**assignment**” with “task”. Go to line 25 and delete the “**task**” on that line.

2. [File system management. 30pt]

(1) There is a file named “secret.txt” located on your Linux Ubuntu. Locate that file and write down its content below (hint: use the “find” command. To access system folders, you need the root access. Type “**sudo**” before your command to exercise root privilege.) [10pt].

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What is the Linux command that you used to locate the file?

Sudo find / -name “secret.txt”

(2) There is a certificate file named “fu.cert” on your system. However, the system administrator forgot where it is used. Search the “/etc” folder and find out where “fu.cert” is used. (hint: use command “grep”) [10pt]

Which file contains keyword “fu.cert”?

/etc/apache2/ssl

What is the Linux command that you used?

i did sudo find / -name “fu.cert” in my \$ pathway.

(3) If you visit “http://localhost/webmail” in Firefox, you could actually see the list of files in that folder (the URL is mapped to /var/www/webmail). What is the Linux command for **forbidding** the Apache server to display directory contents of “webmail” and all of its nested subfolders? (hint: chmod) [10pt]

You /var/www then you sudo chmod 711 webmail because you dont want them to read.

3. [Basic Networking Admin. 50pt]

(1) List all of the IP addresses associated with your Linux Ubuntu image. [5pt]

- 169.254.236.100

- 10.0.3.15

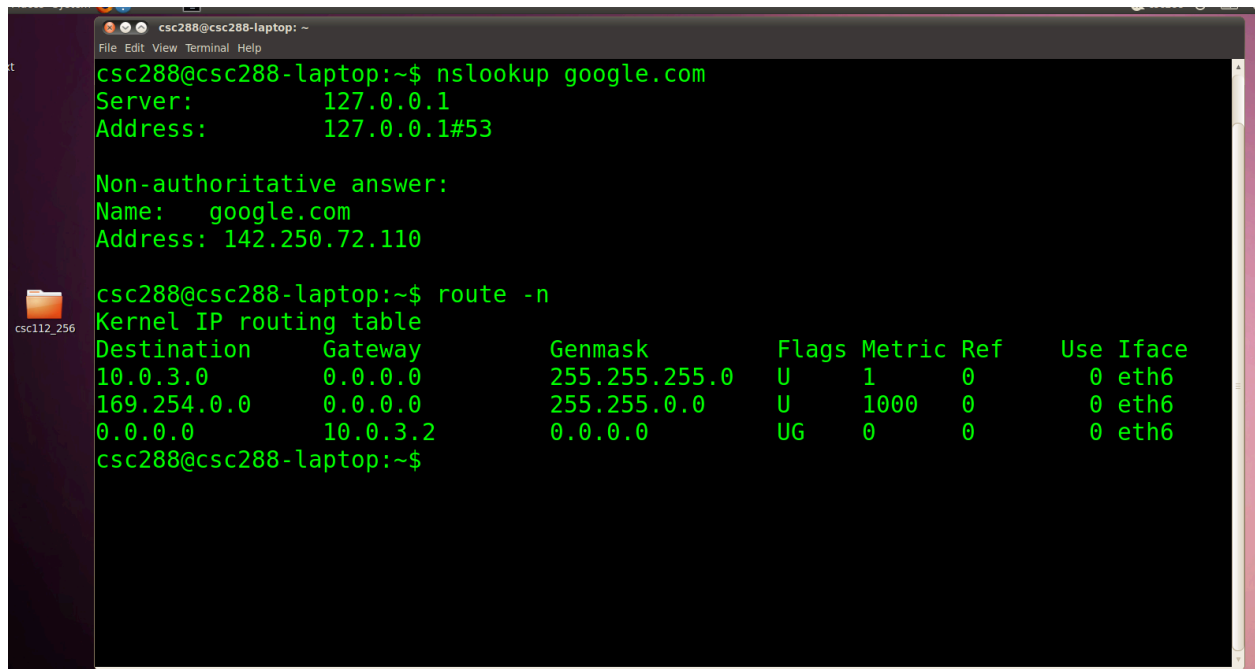
- 127.0.0.1

(2) List the name and MAC address of all of the network adapters on your Linux Ubuntu image. [5pt]

08:00:27:f9:63:01

08:00:27:f9:63:02

(3) Study the routing table (using “route -n”). For an outgoing packet to www.google.com, which adaptor does it go through? [5pt]



The screenshot shows a terminal window with the following output:

```
csc288@csc288-laptop:~$ nslookup google.com
Server:      127.0.0.1
Address:     127.0.0.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.72.110

csc288@csc288-laptop:~$ route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
10.0.3.0    0.0.0.0 255.255.255.0 U 1 0 0 eth6
169.254.0.0 0.0.0.0 255.255.0.0 U 1000 0 0 eth6
0.0.0.0     10.0.3.2 0.0.0.0 UG 0 0 0 eth6
csc288@csc288-laptop:~$
```

eth6

(4) List all of the open ports (with both the numerical value and its well known name) on your Linux Ubuntu image. What is the Linux command for doing so? [10pt]

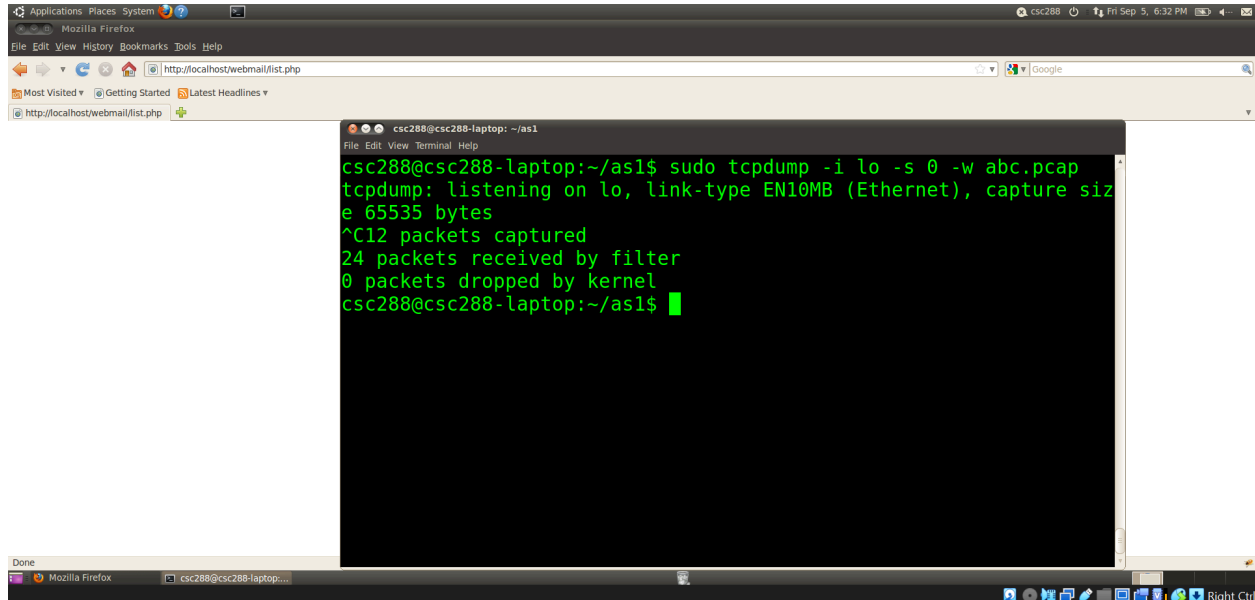
22/tcp (ssh) — sshd
23/tcp (telnet) — inetd
53/tcp (domain/DNS) — named
80/tcp (http/www) — apache2
443/tcp (https) — apache2
631/tcp (ipp) — CUPS cupsd (localhost only)
3306/tcp (mysql) — mysqld (localhost only)
8651/tcp (gmetad) — gmetad
8652/tcp (gmetad) — gmetad
953/tcp (rndc – BIND control) — named (localhost only)
53/udp (domain/DNS) — named
68/udp (bootpc – DHCP client) — dhclient
5353/udp (mdns) — avahi-daemon
44051/udp (ephemeral/avahi-daemon)

Sudo Isot -i

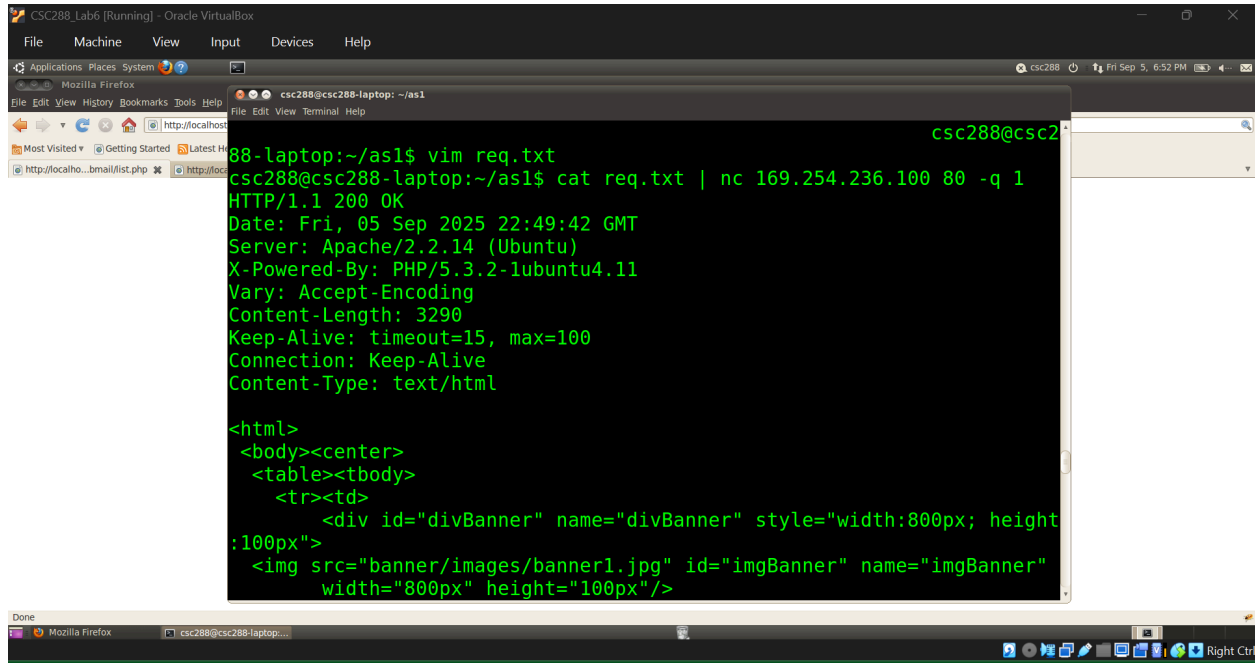
(5) Use a Linux command (without the use of firefox) to download HTML source of “http://localhost/webmail/login.php”. What is the Linux command that you used? [10pt]

wget -O login.html "http://localhost/webmail/login.php"

(6) Visit the webmail system located at http://localhost/webmail using Firefox. Then repeat the log-in process and use TCPDUMP to record all the traffic data and save it to “abc.pcap”. [5pt] (hint: use “stu1” as username and “abc123” as password)



(7) Copy “abc.pcap” to “req.txt” and modify it, so that you could replay the log-in session [hint: use netcat “nc” to replay the HTTP request]. Upload req.txt to Blackboard. [10pt]



```
csc288@csc288-laptop: ~/as1
csc288@csc288-laptop:~/as1$ vim req.txt
csc288@csc288-laptop:~/as1$ cat req.txt | nc 169.254.236.100 80 -q 1
HTTP/1.1 200 OK
Date: Fri, 05 Sep 2025 22:49:42 GMT
Server: Apache/2.2.14 (Ubuntu)
X-Powered-By: PHP/5.3.2-1ubuntu4.11
Vary: Accept-Encoding
Content-Length: 3290
Keep-Alive: timeout=15, max=100
Connection: Keep-Alive
Content-Type: text/html

<html>
<body><center>
<table><tbody>
<tr><td>
<div id="divBanner" name="divBanner" style="width:800px; height:100px">

```

4. Process management [10pt]

(1) What is the command for restarting the Apache service? [5pt]

Sudo systemctl restart apache2

(2) How many Apache services are running on the Linux image? Show the command that you used. [5pt]

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pgrep -c apache2

----- * BONUS * -----

5. Python Script [25pt]

During a security contest, to prevent your opponent from gaining the control of your system, you have to ensure the integrity of several important system files, e.g., `/etc/passwd`. Write a Python script which will be executed when system boots. It will first make a copy of the `/etc/passwd` file and periodically restores `/etc/passwd` from the backup copy. Name your file as **`savepasswd.py`**. You have to describe below how you make sure that the script is run at boot and prove that it is indeed running (and make periodical restoration) with proper screenshots.

```
File Edit View Terminal Help
csc288@csc288-laptop:~$ cd as1/
csc288@csc288-laptop:~/as1$ ls
req.txt  savedpasswd.py  vim.txt
csc288@csc288-laptop:~/as1$ vim savedpasswd.py
csc288@csc288-laptop:~/as1$ chmod +x savedpasswd.py

csc288@csc288-laptop:~/as1$
```

```
File Edit View Terminal Help
csc288@csc288-laptop:~$ ps aux | grep saved
root      2588  0.1  0.3   7628  3528 pts/0    S
   21:58   0:00 python savedpasswd.py
csc288    2613  0.0  0.0   3320   784 pts/1    R+
   21:59   0:00 grep --color=auto saved
csc288@csc288-laptop:~$
```

```
csc288@csc288-laptop:~$ ps aux | grep saved
root      2588  0.1  0.3   7628  3528 pts/0    S
 21:58    0:00 python savedpasswd.py
csc288    2613  0.0  0.0   3320   784 pts/1    R+
 21:59    0:00 grep --color=auto saved
csc288@csc288-laptop:~$ echo "#testchange123" | sudo
tee -a /etc/passwd tail /etc/passwd
#testchange123
csc288@csc288-laptop:~$ tail /etc/passwd
snort:x:115:123:Snort IDS:/var/log/snort:/bin/false
telnetd:x:116:124::/nonexistent:/bin/false
vboxadd:x:999:1::/var/run/vboxadd:/bin/false
jetty:x:117:125::/usr/share/jetty:/bin/false
mysql:x:118:126:MySQL Server,,:/var/lib/mysql:/bin
/false
ganglia:x:119:127:Ganglia Monitor:/var/lib/ganglia:
/bin/false
bind:x:120:128::/var/cache/bind:/bin/false
sshd:x:121:65534::/var/run/sshd:/usr/sbin/nologin
xfu1010:x:1001:1010::/home/xfu1010:/bin/sh
peercast:x:122:129::/usr/share/peercast:/bin/false
csc288@csc288-laptop:~$
```

```
ps acsc288@csc288-laptop:~$ ps aux | grep savedpasswd.py | grep -v grep
root      828  0.0  0.1   2696  1076 ?        Ss   16:42   0:00 /bin/s
h -c /usr/bin/python /home/csc288/as1/savedpasswd.py >> /var/log/savedp
asswd.log 2>&1
root      840  0.1  0.3   7252  3464 ?        S    16:42   0:00 /usr/b
in/python /home/csc288/as1/savedpasswd.py
csc288@csc288-laptop:~$
```

IN SUDO CRONTAB -E

```
File Edit View Terminal Help
# m h dom mon dow   command
@reboot /usr/bin/python /home/csc288/as1/savedpasswd.py >> /var/log/sav
edpasswd.log 2>&1
~
~
~
```